

- Q.30 What are stadia constants? How are they determined?
 Q.31 List the methods of setting out simple circular curves.
 Q.32 Explain the need and use of transition curves in highways.
 Q.33 Explain how a vertical curve is set out.
 Q.34 Write a short note on Total Station and its advantages.
 Q.35 Write any five differences between auto level and dumpy level.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 A simple circular curve has a radius of 800 m. The two tangents intersect at an angle of 120° . The chainage of point of intersection in 1000 m. Find
 a) Tangent length
 b) Length of curve
 c) Chnlage of beginning of curve
 d) Chainage of point of end of curve
 e) Length of long chord
 f) Degree of curve
- Q.37 With telescope horizontal, the readings taken by upper and lower wires were 1.060 and 3.840 m. Find the horizontal distance and elevation of staff station if the R.L. of instrument axis is 341.540 m. The central hair reading was 2.450. The constant of tacheometer used were $f/i = 100$ $(f+d) = 0.4$ m
- Q.38 Explain
 a) Temporary adjustments of a theodolite (5 marks)
 b) Traversing using included angle method (5 marks)

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4th Semester/ Civil, Constr. Mgmt., Highway Engg. Subject: Surveying-II

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Contour lines are drawn on the map of represent
 a) Trees b) Elevations
 c) Roads d) Building
- Q.2 The instrument used to measure horizontal and vertical angles is:
 a) Compass b) Dumpy level
 c) Theodolite d) Planimeter
- Q.3 The distance between two consecutive contours is called:
 a) Horizontal equivalent b) Vertical interval
 c) Contour interval d) Contour factor
- Q.4 Tachometer is primarily used for :
 a) Leveling
 b) Earth work computation
 c) Angle measurement
 d) Distance measurement

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- Q.5 A vertical curve is used in roads and railways for:
 a) Turning b) Drainage
 c) Slope change d) Speed control
- Q.6 Which of the following is a modern surveying instrument?
 a) Chain b) Compass
 c) Total Station d) Tape
- Q.7 Which of the following methods is used for setting out circular curves?
 a) Chord bisection b) Radiation
 c) Triangulation d) None
- Q.8 Height of an inaccessible object can be determined using:
 a) Plane table b) Compass
 c) EDM d) Theodolite
- Q.9 Transition curves are provided to :
 a) Increase speed
 b) Improve visibility
 c) Provide gradual change in curvature
 d) Reduce cost
- Q.10 Auto level is used to:
 a) Measure angles b) Determine bearings
 c) Find elevations d) Calculate area

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 The purpose of contouring is to represent _____ of the ground surface.

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- Q.12 The axis which lies in the line of sight of the theodolite is _____ axis.
- Q.13 The instrument that combines EDM and theodolite functions is called _____.
- Q.14 Tachometric survey is useful where _____ is not possible.
- Q.15 Contours never _____ each other.
- Q.16 True or False: A simple curve has only one radius.
- Q.17 Vertical curves are mainly of two types – crest and _____.
- Q.18 EDM stands for _____.
- Q.19 _____ is used for measuring area directly from a plan.
- Q.20 True or False : planimeter is a digital instrument for angle measurement.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define contour and explain its uses.
- Q.22 Explain any two methods of contouring.
- Q.23 Write short notes on interpolation of contours.
- Q.24 What are the characteristics of contour lines? (any five)
- Q.25 Describe the procedure of prolonging a straight line using a theodolite.
- Q.26 What is face left and face right observation in theodolite surveying?
- Q.27 Differentiate between included and deflection angles.
- Q.28 Describe any two errors in theodolite survey and how to reduce them.
- Q.29 Write the principle of stadia tachometry.

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